

Oladipupo Ridwan Bello, PhD

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SUMMARY

Geneticist and computational immunologist translating single-cell multi-omics and structural biology data into predictive and generative machine learning models of the immune system. Dedicated to accelerating the discovery and development of broad-spectrum and targeted immunotherapies for human and animal health.

EDUCATION

PhD in Genetics and Genomics

Department of Animal Sciences, University of Maryland, College Park, USA, 2025

Dissertation: Exploring the Drivers of Public T-Cell Receptors Using Deep Learning and Transcriptomics

Credential Verification: <https://registrar.umd.edu/ceValidate.html> (CeDiD: 267W-D70Z-ONOW)

MSc in Breeding & Genetics (Distinction)

Department of Animal Science, University of Ibadan, Nigeria, 2019

BSc in Animal Science (First Class Honors)

Department of Animal Science, University of Ibadan, Nigeria, 2015

RELEVANT SKILLS

- Programming: R, Python, Bash, SQL, C, Rust.
- Bioinformatics: Bulk/single-cell RNA-Seq, WGS, ChIP-Seq, ATAC-Seq, TCR/BCR repertoire analysis, structural modeling, metagenome analysis, immune GWAS.
- Statistical Analysis: Hypothesis testing, linear and generalized linear models, mixed models, survival analysis, multivariate statistics, time series analysis, Bayesian statistics.
- Machine Learning: Neural networks, random forest, transformers, generative models, classification, regression, clustering.
- Laboratory Techniques: Tissue Dissection, Microscopy, DNA Extraction, PCR, Gel Electrophoresis, Molecular Cloning.
- Environments and Tools: Git, Docker, Singularity, HPC/Slurm, Quarto, Linux, macOS, Windows.

RESEARCH EXPERIENCE

Doctoral Research Assistant (January 2021 - December 2025)

Department of Animal Sciences, University of Maryland, College Park, USA

- *Identified molecular and structural drivers of public T cell receptors in human and bovine immunity*
 - Distinguished public from private T cell receptor nucleotide sequences across millions of records by building a convolutional neural network classifier achieving over 85% precision and recall.
 - Revealed structural features associated with receptor sharing across individuals by modeling 3-dimensional T-cell receptor structures in MHC contexts.
 - Uncovered distinct cytotoxic effector phenotypes in public T cells by profiling their transcriptomes from single-cell RNA-Seq data.
 - Mapped intercellular communication networks between public/private T cell subsets and other immune cell populations.
 - Identified candidate genes associated with T cell receptor publicness through differential expression and functional enrichment analyses.
 - Identified candidate regulatory genes linking public T-cell receptor signatures to complex traits through GWAS summary statistics and functional pathway enrichment.
- *Characterized the relationship between gut microbial diversity and public T cell receptor diversity in surrounding tissues.*
- *Identified genetic and epigenetic mechanisms underlying resistance to virally-induced T cell lymphoma using an avian cancer model*
 - Identified candidate resistance genes by comparing gene expression profiles of lymphoma-resistant and susceptible avian model lines with bulk RNA-Seq.
 - Validated and refined candidate gene lists using ChIP-Seq and ATAC-Seq chromatin accessibility data.
 - Detected RNA editing events implicated in T cell lymphoma resistance by exploiting the highly inbred nature of the models.
 - Characterized biological functions and pathways of co-expressed genes through network analysis.
- Served as Laboratory Compliance Officer, ensuring the lab met university safety and regulatory standards.

TEACHING EXPERIENCE

Graduate Teaching Assistant (January 2021 - December 2025)

Department of Animal Sciences, University of Maryland, College Park, USA

- *ANSC 627/327 (Quantitative and Molecular Genetics)*
 - Led office hours and review sessions for undergraduate and graduate students.
- *ANSC 447 (Physiology of Mammalian Reproduction Lab)*
 - Led laboratory sessions on gross anatomy and histology of mammalian reproductive systems.

Tutorial Assistant (April 2017 - September 2018)

Department of Animal Science, University of Ibadan, Nigeria

- Led tutorials on molecular/quantitative genetics and introductory statistics for undergraduate students.

PUBLICATIONS

Bello, O.R. (2026). *R Lab: Data Science Env for R in VS Code* [Software]. VS Code Marketplace. [\[URL\]](#)

Bello, O.R., Arvind, J., & Johnson, P.L.F. (In Review). Intrinsic Determinants of TCR Publicness and Their Structural Implications. *PLOS Computational Biology*.

Bello, O.R. & Johnson, P.L.F. (In Review). Distinct Cytotoxic Effector Phenotypes of Public T Cells. *Frontiers in Immunology*.

Bello, O.R., & Johnson, P.L.F. (January 10-15, 2025). *Public T Cell Receptors in Bovine Immunity*. International Plant & Animal Genome Conference, San Diego, California, USA. [\[URL\]](#)

Bello, O.R., Chu, Q., & Song, J. (July 10-13, 2023). *Temporal profiling of the bursa transcriptome reveals systemic differences induced by Marek's disease virus*. PSA, Philadelphia, Pennsylvania, USA. [\[URL\]](#)

Ewuola, E.O., Adeyemi, A.A., & **Bello, O.R.** (2020). Variations in haematological & serum biochemical indices among White Fulani bulls, Ouda rams & West African Dwarf bucks. *NJAP*, 44(1), 136-143. [\[URL\]](#)

SELECTED HONORS AND AWARDS

- Invited Peer Reviewer (2025): Elsevier
- Jacob K. Goldhaber Travel Grant (2025): University of Maryland, College Park, USA
- Animal Health & Care Academy Fellow (2024/2025): MANRRS, USA (sponsored by Merck & Zoetis)
- Dean's Fellowship (2021): Graduate School, University of Maryland, College Park, USA
- EducationUSA Opportunity Funds Program Fellow (2019): United States Embassy in Nigeria
- University of Ibadan Master's Scholarship & Tutorial Assistantship (2017): University of Ibadan, Nigeria
- Overall Best (Nationwide), Graduate Animal Scientist Exam (2016): Nigerian Institute of Animal Science